

Influence of water and food consumption on inadvertent antibiotics intake among general population.



Antibiotic entry into the water environment has been of growing concern. However, few investigations have been performed to examine the potential for indirect human exposure to environmental antibiotic residues. We evaluated the contribution of drinking water and major food consumption to inadvertent intake of antibiotic residues among general human population in Korea. We estimated daily human intake of six antibiotics, i.e., sulfamethazine (SMZ), sulfamethoxazole (SMX), sulfathiazole (STZ), trimethoprim (TMP), enrofloxacin (EFX), and roxithromycin (RTM), by measuring the concentrations of the antibiotics and their major metabolites in urine from general population in Korea (n=541). In addition, we measured antibiotics from source water of drinking water as well as in tap water samples, and surveyed water consumption rates among the study population. To assess the contribution of dietary factor, we also surveyed consumption pattern for several major foods which are suspected of antibiotics residue. SMZ, Sulfamethazine-N4-acetyl (SMZ-N4), TMP, EFX, ciprofloxacin (CFX), and RTM were detected up to 448, 6210, 11,900, 6970, 32,400, and 151pg/ml in the urine samples, respectively. Estimates of daily intake of major antibiotics did not appear to be related with consumption of drinking water although antibiotics were frequently detected in source waters (10-67ng/l). Consumption of several foods correlated significantly with urinary excretion of several antibiotics. Daily intake estimates of EFX and CFX were associated with consumption of beef, pork, and dairy products; those of SMZ and TMP associated with pork and dairy products; and that of TMP related with raw fish. Daily antibiotics intake estimates however did not exceed the acceptable daily intake levels. Copyright 2010 Elsevier Inc. All rights reserved.